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DAA LAB

LAB EXPERIMENTS

EXPERIMENT – 8

CLOSEST PAIR OF POINTS - BRUTE FORCE

AIM:

To find the closest pair of points in a two-dimensional plane using the brute-force method and analyze its time complexity.

ALGORITHM:

1. Start the program.
2. Initialize a list of points with their x and y coordinates.
3. Compare every pair of points using nested loops.
4. Calculate the Euclidean distance between each pair.
5. Update the minimum distance and closest pair when a smaller distance is found.
6. Display the closest pair and minimum distance.
7. Stop the program.

PROCEDURE:

1. Store the coordinates of the given points in a list.
2. Use nested loops to select every possible pair.
3. Calculate the distance using the Euclidean distance formula:
4. Compare each calculated distance with the minimum distance.
5. Store the pair with the smallest distance.
6. Print the closest pair, distance, and time complexity.

PYTHON CODE:

```python
import math
points = [(2, 3), (12, 30), (40, 50),
          (5, 1), (3, 4)]
min_dist = float('inf')
closest = None
for i in range(len(points)):
    for j in range(i + 1, len(points)):
        x1, y1 = points[i]
        x2, y2 = points[j]
        d = math.sqrt((x2-x1)**2 + (y2-y1)**2)
        if d < min_dist:
            min_dist = d
            closest = (points[i], points[j])
print("Closest pair:", closest)
print("Minimum distance:", round(min_dist, 2))
print("Time Complexity: O(n^2)")
```

SAMPLE OUTPUT:

```
Output

Closest pair: ((2, 3), (3, 4))
Minimum distance: 1.41
Time Complexity: O(n^2)
```

RESULT:

The closest pair was found successfully with a minimum distance of 1.41 units and time complexity (O(n^2)).